Market Power and the Gender Career Gap

Evidence from Matched HR Records in Chile

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Wages are only one dimension of how careers unfold

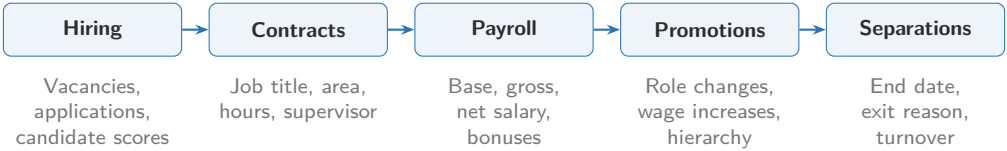
- Gender gaps persist despite convergence in education and declining fertility (Goldin, 2014; Blau & Kahn, 2017). Most research focuses on *wages*, but firms also decide **who gets hired** and **who gets promoted**.
- Search frictions give firms market power over workers (Manning, 2003). Women face fewer outside options than men (Webber, 2016; Caldwell & Danieli, 2022). **This depresses wages — but does it also distort hiring and promotion decisions?**
- **The mechanism:** A firm deciding whether to promote weighs the output gain against the wage increase — but also *retention*. If a male worker has many outside options, the firm promotes to keep him. If a female worker has fewer options, the firm can delay at little cost.
- **This paper:** connects imperfect competition to gender gaps in career dynamics at *both* the hiring and promotion margins, using matched employer-employee-applicant data from Chile.

Chile: convergence in education and fertility, persistence in gaps

- Women surpassed men in tertiary attainment (45% vs. 37%), TFR fell to 1.03 — yet the gender wage gap persists at 15–20% and female labor force participation remains at 53% (OECD avg: 65%)
- These gaps have become a central policy concern, prompting a series of **labor market reforms** during the 2022–2026 period:
  - **40-hour workweek** (Ley 21.561, phased 2024–2028) — changes the premium on long-hours availability
  - **Flexible work for caregivers** (Ley 21.645, Jan 2024) — parents of children under 14 can request remote/flexible work
  - **Childcare mandate** (Art. 203) — firms with $\geq$20 female workers must fund childcare; firms bunch at 19
- Each reform shifts a different **parameter of the firm's problem** and can be exploited for identification

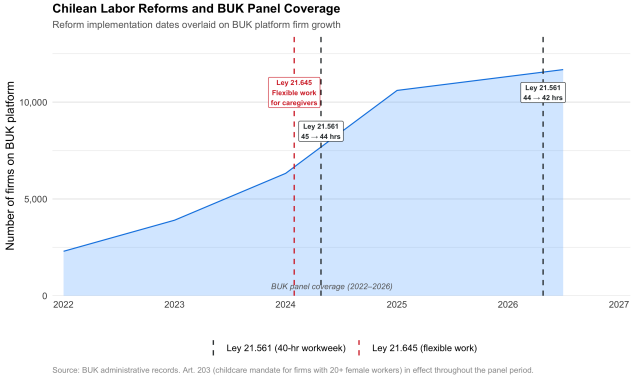
BUK records the full employment relationship

BUK is an HR and payroll platform used by 19,000+ Chilean firms across all 20 ISIC sectors. It records each stage of the employment relationship:



- **Worker–month panel:** 4.7M workers, 8.3M job spells, with firm, occupation, and supervisor identifiers
- **Recruitment module:** 186K selection processes, 2.7M applications — observe the hiring pipeline from application through scoring
- **Complemented by** the Work in Progress survey (perceptions, satisfaction, preferences) ▸ Survey details

Panel coverage and reform timeline (2022–2026)



- Platform grew from ∼6,500 firms (2022) to 19,000+ (2026); rapid adoption, not firm entry
- Three reforms fall **within the panel window**, creating pre/post variation
- Art. 203 childcare mandate in effect throughout — cross-sectional RD at 20-female-worker threshold

Firms: 19,000+ Chilean firms, skewed toward large employers

	N firms		% of workers
Large (>250 employees)	5,459	(28.4%)	79.8%
Medium (50–250)	6,127	(31.9%)	14.5%
Small (<50)	7,612	(39.6%)	5.7%
Total		**19,358**	**100%**

- 20 SII economic sectors represented; top 5: retail, professional services, admin services, finance, manufacturing
- Linkable to SII tax records for 87% of firms (sector, sales bracket, equity range)
- Firm identifiers allow linking across all tables (workforce, salary, assignments, recruitment)

Workers: 4.7M employees with detailed job spells

Unique workers	4,712,667
Job spells	8,273,604
Male / Female	62.8% / 37.2%
Job spells per worker	
Mean	5.86
Median	3.00
P10 / P90	1.00 / 14.00

- **Hierarchy:** supervisor identifier (`boss_id`) enables org structure and promotion measurement
- **Job characteristics:** area, standardized occupation (CIUO-08 crosswalk), contract type (fixed-term / indefinite), workday type (full-time, part-time, Art. 22 exempt)
- **Separations:** end date for 83.2% of completed spells; exit reason (voluntary / involuntary)

Compensation: monthly payroll and individual bonus assignments

Two complementary tables allow decomposition of total compensation:

- **Monthly payroll** (`adm_salary`): base, gross, and net salary for each worker-month
  - Taxable and non-taxable earnings observed separately
  - Liquidation type distinguishes regular salary, severance (*finiquito*), vacation pay, and bonuses
  - Allows fixed vs. variable compensation decomposition
- **Bonus assignments** (`adm_assign`): individual assignment-level records
  - Categories: performance bonus, tenure bonus, commission, *aguinaldo*, transportation, meal allowance
  - Each record has start/end dates and monetary amount
  - Tracks whether the firm rewards through *promotions* (title + base wage) or *bonuses* (variable pay without advancement)

Recruitment: 186K hiring processes with application-level data

Selection processes	186,457
Applications	2,725,982
Unique candidates	839,664
Median apps per vacancy	2
Mean apps per vacancy	18.7
P90 apps per vacancy	42
Candidate gender: Male / Female	56.0% / 43.0%

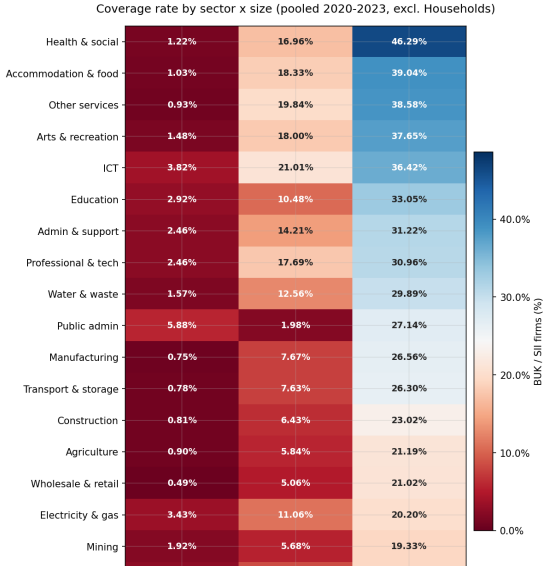
- Each vacancy records functional area, salary range offered, and reason (new position, replacement, temporary)
- Each application has candidate demographics, referral source (portals 40.5%, company 23.7%, job boards 14.7%), and application score
- Heavy right tail — some vacancies attract 1,000+ applicants

Sample selection: BUK vs. the universe of formal Chilean firms

	Firms (%)		Workers (%)	
Size	BUK	SII	BUK	SII
Micro/Small	39.6	95.3	5.7	29.9
Medium	31.9	3.2	14.5	16.5
Large	28.5	1.5	79.8	53.6

- Large firms over-represented — but this is an **advantage**: model predictions on hiring screening and promotion thresholds are most testable in firms with structured pipelines
- **Gender:** BUK sample is 37.2% female; formal employment is ∼42% female — modest under-representation
- **Sectors:** public administration absent (public employers don't use platform); manufacturing under-represented; finance, education, real estate over-represented

Coverage varies by sector and firm size



Coverage rate = BUK firms / SII firms, by ISIC sector × firm size.

- Strong coverage among **large firms** (30–46%)
- Thin in micro/small and public administration
- Platform expanding into smaller firms (avg. empl./firm: 257 in 2022 → 166 in 2026)

Many possible research questions

1. **Monopsony power and career dynamics**
   - Do firms with greater market power have larger gender gaps in promotions and hiring screening?
2. **Bonuses vs. promotions as retention tools**
   - Do firms substitute variable pay (performance bonuses, commissions) for career advancement, especially for women?
3. **The 40-hour workweek and flexible schedule workers**
   - Does restricting Art. 22 exemptions hurt workers who valued schedule flexibility — particularly mothers?
4. **The broken rung at first promotion**
   - Is there a gender gap at the first promotion, and do long-hours norms explain who gets promoted?
5. **Wage growth over the life cycle**
   - How does market power shape wage trajectories by gender at different career stages?

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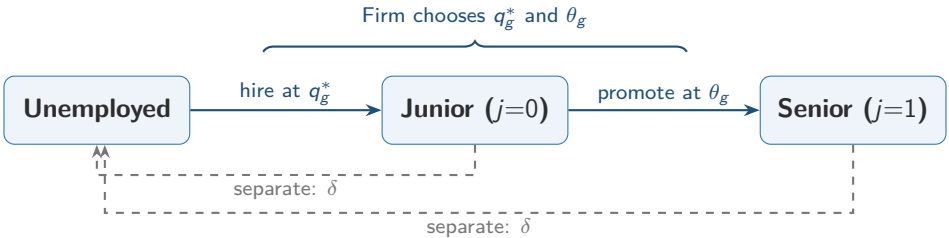
Three literatures have developed in parallel

- **Imperfect competition in labor markets:** search frictions give firms market power (Manning, 2003); on-the-job search models show outside options shape the employment relationship (Burdett & Mortensen, 1998; Postel-Vinay & Robin, 2002); labor supply elasticities to firms are low (Sokolova & Sorensen, 2021) — **but the focus is on wages**
- **Outside options and gender:** women exhibit lower labor supply elasticities (Webber, 2016; Sharma, 2023); differences in outside options driven by commuting and mobility explain a meaningful share of the gender wage gap (Caldwell & Danieli, 2022; Morchio & Moser, 2024); women capture a smaller share of firm pay premiums (Card et al., 2016) — **but still only studied for *wages***
- **Gender and career dynamics:** subjective "potential" assessments account for half the promotion gap (Benson et al., 2026); employers screen applicants differently by gender (Bertrand & Mullainathan, 2004; Kline et al., 2022) — **but no connection to *market structure***
- **This paper:** connects imperfect competition to gender gaps in career dynamics at both hiring and promotion margins

Model: on-the-job search with hiring and promotion

Building on: Burdett & Mortensen (1998), extended with promotion decisions (Gibbons & Waldman, 1999) and gender-specific outside options (Sharma, 2023).



- Two worker types ($g \in \{M, F\}$), firms with two job levels. Production is **gender-neutral** — gender enters entirely through outside options
- **Key parameter:** outside offer arrival rate λ_g, with $\lambda_F < \lambda_M$
- Firms choose a **screening threshold** at hiring and a **promotion threshold** — both shape careers

The promotion mechanism: why monopsony distorts promotions

The firm's promotion decision balances **two forces**:

1. **Direct profit effect:** promoting raises output ($a_1 > a_0$), but also raises the wage ($w_{g1} > w_{g0}$). This trade-off is gender-neutral.

2. **Retention effect:** promoting makes the worker less likely to leave. This benefit depends on λ_g:

Men (high λ_M):

- "If I don't promote him, he'll leave"
- Retention urgency is **high**

$\Rightarrow$ Promote to retain

Women (low λ_F):

- "Even without promotion, she won't leave"
- Retention urgency is **low**

$\Rightarrow$ Delay is low-cost

$\Rightarrow \theta_F^* > \theta_M^*$: higher bar for women, not from productivity differences, but because firms face **less competitive pressure** to promote them.

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The hiring mechanism: why monopsony distorts screening

When $\lambda_F < \lambda_M$, two **opposing forces** shape the firm's screening threshold q_g^*:

1. **"Longer tenure" force** $\rightarrow$ screen women *less*
   Women stay longer at the firm $\rightarrow$ each hire generates surplus over more periods $\rightarrow$ the expected value of a marginal hire is higher $\rightarrow$ firm should lower the bar to hire more women

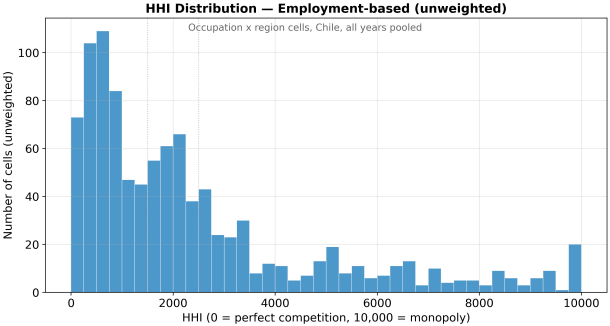
2. **"Less competition" force** $\rightarrow$ screen women *more*
   Women have fewer outside options $\rightarrow$ they apply even when the bar is high $\rightarrow$ firm can raise q_F^* without losing its applicant pool $\rightarrow$ "cherry-pick"

The sign of $q_F^* - q_M^*$ is ambiguous — which force dominates is an **empirical question**.

Test: BUK application-to-hire conversion rates by gender, across markets with different concentration.

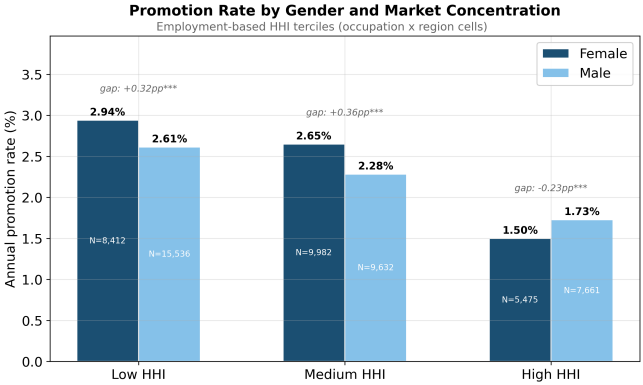
Market concentration varies substantially across labor markets

HHI computed over occupation $\times$ region cells (8 functional areas $\times$ 16 regions $=$ 1,017 cells):



- Employment-based mean HHI $=$ 2,249 (moderately concentrated), but substantial right tail
- Geographic variation: Santiago (HHI $\approx$ 209) vs. peripheral regions like Aysén (HHI $\approx$ 5,690)

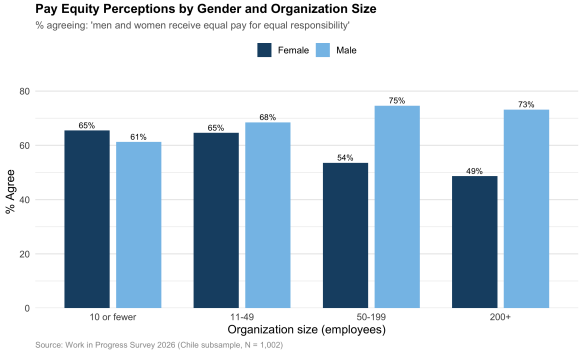
The gender promotion gap reverses with market concentration



Source: BUK admin data (Chile, 2020-2026). Definition: role ID change + salary >5%.

- Aggregate promotion rates nearly identical (F: 2.10%, M: 2.04%; 73K promotions over 3.5M person-years)
- But **the gap reverses with concentration**: women promoted more in competitive markets (+0.32pp), less in concentrated markets (−0.23pp)

Survey evidence: perceived inequity widens with organization size



- Women perceive **less pay equity** than men, and the gap **widens with organization size** — paralleling the HHI interaction in the promotion data
- Women report fewer promotion opportunities and face worse outcomes at every stage of salary negotiation (WiP Survey 2026)
- Lack of growth is the **top reason** workers of both genders cite for wanting to change jobs

Remaining descriptive work and next steps

Several pieces of evidence that would strengthen the case are still in progress:

- **Job-to-job transition rates by gender** — the most direct test of whether women face fewer outside options. Requires supplementary administrative records (unemployment insurance data)
- **Application-to-hire conversion rates by gender $\times$ HHI** — tests the model's hiring prediction; the hired variable in the recruitment module is still being constructed
- **Wage growth at promotion by gender $\times$ HHI** — tests whether the promotion gap translates into different wage trajectories
- **Structural labor supply elasticity estimates** — complement HHI with estimates that distinguish vertical and horizontal sources of monopsony power (Sharma, 2023; Schubert et al., 2024)

Formal model: workers' problem

Workers of gender g are either **unemployed** or **employed** at level $j \in \{0, 1\}$ in a firm of type p.

Value of unemployment:

$$rU_g = b + \lambda_g^u \int \max\{V_{g0}(w) - U_g,\ 0\}\, dF_g(w)$$

Value of employment at level j:

$$\begin{aligned} rV_{gj}(p) = w_{gj}(p) + \lambda_g \int \max\{V_{g0}(w') - V_{gj}(p),\ 0\}\, dF_g(w') + \delta[U_g - V_{gj}(p)] \\ + \mathbb{1}_{j=0} \cdot \mu_g(p)[V_{g1}(p) - V_{g0}(p)] \end{aligned}$$

- λ_g: on-the-job outside offer arrival rate ($\lambda_F < \lambda_M$)
- $\mu_g(p)$: promotion rate at firm p for gender g (determined by firm's choice)

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Formal model: firm's problem

The firm chooses $(w_{g0}, w_{g1}, q_g^*, \theta_g)$ to maximize steady-state profits:

$$\pi_g(p) = \sum_{j \in \{0,1\}} \left[y_j(p) - w_{gj} \right] \cdot n_{gj}(p)$$

where the steady-state workforce $n_{gj}(p)$ depends on three flows:

1. **Inflow** (hiring): $h_g(p) = \lambda_g^u \cdot u_g \cdot [1 - H(q_g^*)] \cdot \mathbb{1}\{V_{g0}(p) \geq U_g\}$
2. **Internal flow** (promotion): $\mu_g(p) = 1 - G(\theta_g)$
3. **Outflow** (separation): $s_g(p) = \delta + \lambda_g \cdot [1 - F_g(V_{gj}(p))]$

In steady state:

$$n_{g0}(p) = \frac{h_g(p)}{\mu_g(p) + s_g(p)}, \qquad n_{g1}(p) = \frac{\mu_g(p) \cdot h_g(p)}{s_g(p) \cdot [\mu_g(p) + s_g(p)]}$$

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Formal promotion condition (two-period simplification)

Setup: Period 1, worker is junior (profit $\pi_0 = pa_0 - w_{g0}$). Between periods, outside offer arrives with probability λ_g. If promoted, continuation value rises: $V_{1g} > V_{0g}$.

Retention probabilities:

$$r_g^P = 1 - \lambda_g[1 - F_g(V_{1g})] \quad \text{(if promoted)}$$

$$r_g^N = 1 - \lambda_g[1 - F_g(V_{0g})] \quad \text{(if not promoted)}$$

Since $V_{1g} > V_{0g}$: $r_g^P > r_g^N$ (promoted workers are easier to retain).

Promote if period-2 expected profit is higher with promotion:

$$\underbrace{(\pi_1 - \pi_0) \cdot r_g^N}_{\text{Direct profit effect}} + \underbrace{\pi_1 \cdot (r_g^P - r_g^N)}_{\substack{\text{Retention value} \\ \text{of promoting}}} \geq 0$$

The retention term is proportional to λ_g: $\quad r_g^P - r_g^N = \lambda_g \left[F_g(V_{1g}) - F_g(V_{0g})\right]$

- As $\lambda_g \to 0$: retention term vanishes $\to$ promote only if $\pi_1 > \pi_0$
- As $\lambda_g \to \infty$: retention dominates $\to$ always promote to retain
- Since $\lambda_F < \lambda_M$: retention motive is weaker for women $\to \theta_F^* > \theta_M^*$

Work in Progress 2026 Survey

- **Sample:** 6,789 respondents in Chile, Colombia, Mexico, Peru
- **Open survey** — anyone can respond; **cannot be linked** to platform data
- **Topics covered:**
  - Demographics, salary, job tenure, firm size
  - Job satisfaction, burnout, work-life balance
  - Gender equity perceptions, discrimination, D&I policies
  - Benefits, performance evaluation, retention intentions
- Provides **subjective** complement to BUK's **administrative** records

Source: BUK Work in Progress Survey 2026.

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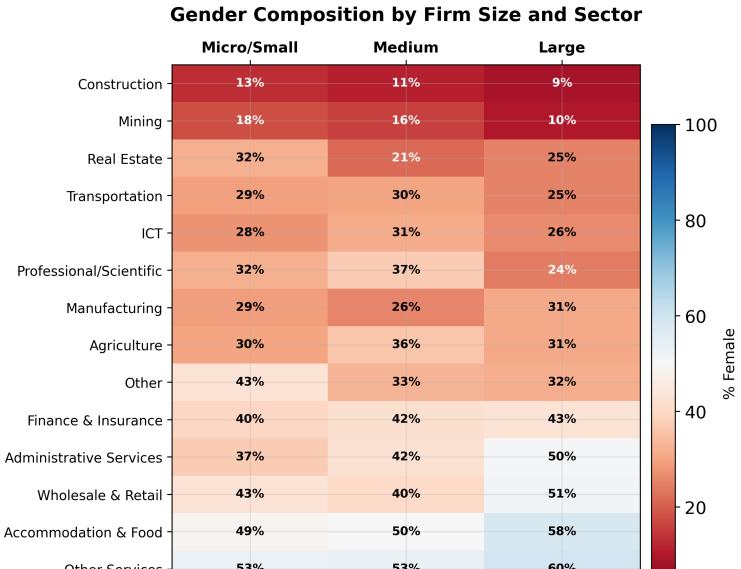
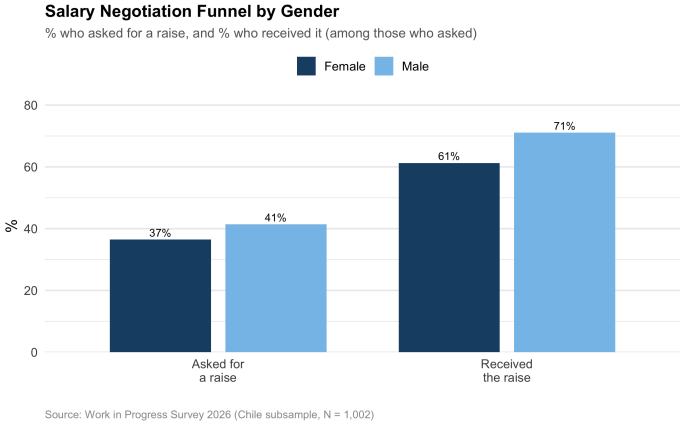
Gender composition of BUK firms by sector and size

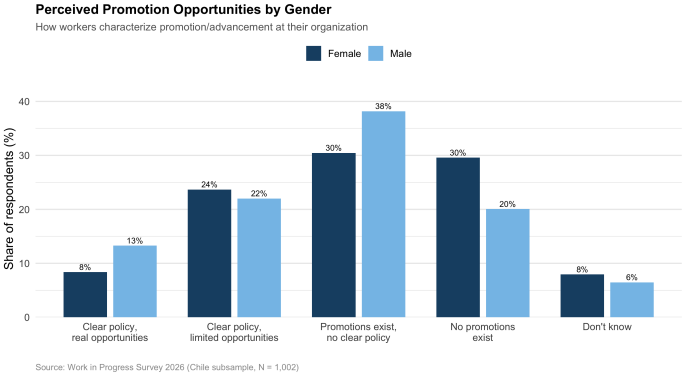


Salary negotiation funnel by gender



Women are less likely to ask for a raise, and conditional on asking, less likely to receive one. Source: WiP Survey 2026, Chile subsample (N = 1,002).

Perceived promotion opportunities by gender



Women report fewer promotion opportunities at their organization. Source: WiP Survey 2026, Chile subsample (N = 1,002).